

Prevalence of High-Fructose Corn Syrup in the U.S. Branded Food Supply: A Cross-Sectional Analysis of the
USDA Global Branded Food Products Database

LIBERTY
UNIVERSITY

Name of Student(s)

Abstract

Background: High-fructose corn syrup (HFCS) is a widely used caloric sweetener in the U.S. food supply, but product-level data on its prevalence are limited. Ingredient lists do not specify amounts, and nutrient databases do not distinguish among types of added sugars.

Objectives: To estimate the prevalence of HFCS as a listed ingredient in U.S. branded food products and to establish a reproducible baseline for future surveillance.

Methods: We conducted a retrospective, cross-sectional, descriptive analysis of the USDA FoodData Central Global Branded Food Products Database (GBFPD), covering products published between April 13, 2019, and December 13, 2025. The primary outcome was the proportion of eligible products listing HFCS as an ingredient. We used descriptive statistics (counts and percentages).

Results: Of 454,324 products identified, 447,607 were eligible for analysis. HFCS was listed as an ingredient in 29,503 products (6.6%), or about 1 in 15. The remaining 418,104 products (93.4%) did not list HFCS.

Conclusions: HFCS remains present in roughly 30,000 U.S. branded food products. Because we measured prevalence by product count rather than weighted by sales or intake, and could not quantify HFCS per product, these findings do not directly measure population exposure. They provide a reproducible baseline for monitoring reformulation and market change. Future work should link ingredient data to sales-weighted datasets and stratify by food category and trade channel.

Introduction

- HFCS is a widely used caloric sweetener, especially in soft drinks, baked goods, ready-to-eat meals, and condiments.
- Excess added sugar is linked to obesity, type 2 diabetes, and cardiometabolic disease. The 2025–2030 Dietary Guidelines state that no amount of added sugars is recommended as part of a healthy diet.
- Product-level HFCS data are limited: ingredient lists give no amounts, and nutrient databases do not separate types of added sugars.

Research question

What proportion of U.S. branded food products list HFCS as an ingredient?

Objectives

- Estimate HFCS prevalence in U.S. branded food products
- Establish a reproducible baseline for future surveillance

Methods

- **Design:** retrospective, cross-sectional, descriptive
- **Data:** USDA FoodData Central Global Branded Food Products Database (GBFPD); 400,000+ industry-supplied products, updated monthly
- **Search window:** April 13, 2019 – December 13, 2025 (FDC published date)
- **Trade channels:** grocery, convenience, mass merchandising, food service, vending, online, military, child nutrition programs, none
- **Excluded:** “Drug” category and New Zealand market (n = 6,717)
- **HFCS search:** ingredients containing “High” AND “Fructose” AND “Corn” AND “Syrup”
- **Analysis:** % of eligible products listing HFCS (Excel)

Results & Conclusions

Results

- **447,607** U.S. branded food products were eligible (454,324 identified before exclusions)
- **29,503 products (6.6%)** listed HFCS as an ingredient, about **1 in 15**
- **418,104 products (93.4%)** did not list HFCS

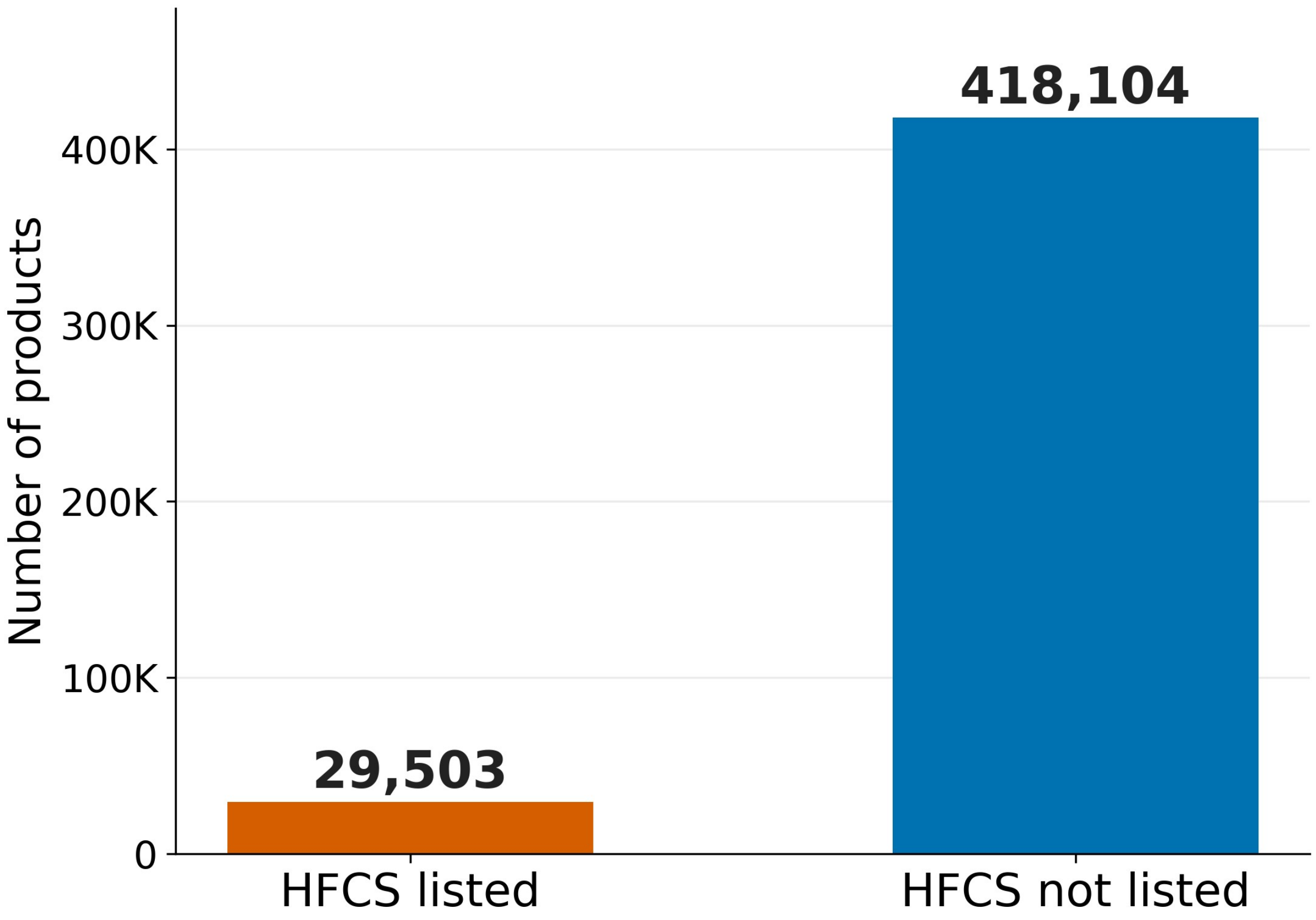
Conclusions

- HFCS is listed in **6.6%** of U.S. branded food products, nearly **30,000** items
- It is a minority ingredient, but still a relevant part of the commercial food supply
- Product counts do not reflect consumption volume or the amount of HFCS per product
- Term-based searching may slightly over- or under-count HFCS products
- This analysis provides a **reproducible baseline** for HFCS surveillance, policy evaluation, and efforts to reduce added sugars

Future Work

1. **Estimate consumer exposure.** Link ingredient data to sales-weighted or scanner data, so prevalence reflects what people buy.
2. **Find where HFCS is concentrated.** Break down prevalence by food category and trade channel.
3. **Strengthen monitoring.** Combine ingredient data with Nutrition Facts added-sugar labeling.
4. **Track change over time.** Repeat the analysis to detect reformulation and market shifts.

Number of U.S. branded food products with and without HFCS



Where is HFCS commonly used?

Food categories where U.S. manufacturers most often use high-fructose corn syrup



HFCS-55 (55% fructose) predominates in soft drinks; HFCS-42 is more common in other processed foods

Categories are drawn from the study background. Category-level prevalence was not analyzed in this study. Illustration is stylized and not tied to any specific brand.

